

Sara Katanchian

Fort Thomas, KY 41075
katancsa@mail.uc.edu — 717-315-7379

CAREER OBJECTIVE

Economics PhD candidate specializing in applied macroeconomics, regional economics, and applied econometrics, seeking an academic or research economist position. Research examines how national economic shocks affect consumption across regions, sectors, and income groups, with particular emphasis on spatial dependence and heterogeneous local responses. Experienced in Bayesian and spatial econometric methods, high-frequency panel data, forecasting, teaching, and policy-relevant empirical research.

EDUCATION

PhD Candidate in Economics, University of Cincinnati, Cincinnati, OH **Expected 2026**

Carl H. Lindner College of Business

Dissertation proposal defended: June 2026

Dissertation: *Essays on Spatial Consumption Dynamics and National Shocks*

Supervisor: Olivier Parent

Fields: Applied Macroeconomics, Regional Economics, Econometrics, and Spatial Economics

MS in Economics, University of Tehran, Tehran, Iran **2015**

Ranked 1st in class, GPA: 19.22/20

Thesis: *Studying the Neutrality of Money: Evidence from OPEC Member States*

Supervisor: Mohsen Mehrara

BS in Economics, University of Tabriz, Tabriz, Iran **2013**

Ranked 3rd in class, GPA: 17.96/20

RESEARCH INTERESTS

Applied Macroeconomics; Regional Economics; Spatial Econometrics; Applied Econometrics; Bayesian Econometrics; Consumption Dynamics; National and Regional Economic Shocks; Income Heterogeneity; Policy Analysis; Time Series; High-Frequency Economic Data; Forecasting; Machine Learning.

PUBLICATIONS AND RESEARCH

Katanchian, S., and Parent, O. “Uncovering Local Consumption Dynamics during COVID-19: Evidence from a Spatial Dynamic Factor Model.”

Revise and resubmit, *Journal of the Royal Statistical Society: Series A*. Presented at the 64th Southern Regional Science Association Annual Meeting, 2025.

Katanchian, S., and Parent, O. “The Geography of Consumption: Local Responses to National Shocks.”

Submitted manuscript. Presented at the 65th Southern Regional Science Association Annual Meeting, 2026.

Katanchian, S., and Parent, O. “Income Heterogeneity in Consumption Responses to National Shocks.”

Work in progress.

Katanchian, S., and Parent, O. “Forecasting Regional Consumer Spending: A Comparison of Bayesian Spatial Dynamic Factor Models and Machine Learning.”

Research in progress.

Katanchian, S. “Studying the Neutrality of Money: Evidence from OPEC Member States.”

International Journal of Applied Economics, Vol. 3, Issue 4, 2015.

Mansouri, M., Mohammad Pour, R., and Katanchian, S. “Studying Money Laundering in the Banking Industry: Using Game Theory.”

Presented at the First National Conference on Industrial Economics, University of Tabriz, 2016.

RESEARCH EXPERIENCE

Graduate Researcher, Department of Economics, University of Cincinnati **2022–Present**

- Develop Bayesian Markov Chain Monte Carlo (MCMC) estimators for spatial dynamic factor models with spatially dependent factor loadings to identify latent national consumption shocks and heterogeneous local responses across large weekly U.S. panels.

- Estimate Bayesian dynamic factor VAR models with spatially structured loadings and within- and cross-sector lag dynamics, using Gibbs sampling, posterior simulation, Monte Carlo experiments, and robustness analysis.
- Construct and analyze high-frequency consumption panels spanning more than 1,600 U.S. counties, 50 states and 10 consumption sectors, and state-by-income-quartile data to study regional and distributional responses to national shocks.
- Develop spatio-temporal forecasting frameworks in MATLAB and Python that compare Bayesian spatial dynamic factor forecasts with autoregressive, VAR, spatial VAR, Random Forest, and XGBoost benchmarks using rolling-origin evaluation.

PROFESSIONAL EXPERIENCE

Incoming Instructor, Thomas More University, Crestview Hills, KY Fall 2026

Robert W. Plaster College of Business

- Appointed to teach two undergraduate economics courses during the Fall 2026 semester.
- Prepare course materials, assessments, presentations, digital learning resources, and learning-management-system content.
- Communicate economic reasoning and quantitative concepts to students with varied academic and technical backgrounds.

Instructor, Department of Economics, University of Cincinnati 2023–2026

- Designed and delivered undergraduate courses in Introduction to Microeconomics, Intermediate Microeconomics, and Game Theory.
- Communicated quantitative concepts and economic reasoning to audiences with diverse technical backgrounds.
- Developed assessments, interpreted student-performance data, and adapted instruction based on learning outcomes.
- Managed online and in-person course delivery, grading, student communication, and learning-management systems.

Teaching Assistant, Department of Economics, University of Cincinnati 2022–2025

- Supported graduate-level Macroeconomics and Econometrics and undergraduate Microeconomics and Macroeconomics.
- Assisted with statistical methods, data analysis, quantitative problem solving, grading, and student support.

SELECTED GRADUATE COURSEWORK

Advanced Macroeconomics	Advanced Econometrics
Bayesian Econometrics	Spatial Econometrics
Time Series Data Analysis	Applied Statistical Modeling
Advanced Microeconomics	Foundations of Finance
Machine Learning	

TECHNICAL SKILLS

Econometrics:	Spatial econometrics, spatial dynamic factor models, Bayesian VAR, panel-data econometrics, time-series analysis, spatio-temporal forecasting
Bayesian Methods:	Bayesian Markov Chain Monte Carlo, Gibbs sampling, Bayesian inference, posterior simulation, posterior predictive analysis
Programming:	Python, MATLAB, R, SQL, Stata, Julia, LaTeX, EViews, GeoDa
Machine Learning:	Random Forest, XGBoost, predictive modeling, rolling-origin forecasting

CERTIFICATIONS

Advanced Monetary Economics , Pierpaolo Benigno, Monetary and Banking Research Institute, Tehran, Iran	2015
The Complete SQL Bootcamp: Go from Zero to Hero , Jose Portilla	2024
Julia Programming for Data Science and Machine Learning , Ankit Mistry, Data Science and Machine Learning Academy	2024

AWARDS

Charles Phelps Taft Dissertation Fellowship , University of Cincinnati	2024–2025
PhD Program Admission Grant , Excellence in Graduate Studies, University of Science and Research, Tehran, Iran	2017
Graduate Program Admission Grant , Excellence in Undergraduate Studies, University of Tehran, Tehran, Iran	2013

LANGUAGES

English: Professional proficiency
Persian and Azeri: Native bilingual proficiency

WORK AUTHORIZATION

U.S. permanent resident authorized to work in the United States without employer sponsorship.

REFERENCES

Olivier Parent

Professor of Economics, University of Cincinnati

Phone: 513-556-3941 • Email: parentor@ucmail.uc.edu

Jeffrey Mills

Professor of Economics, University of Cincinnati

Phone: 513-556-6000 • Email: jeffrey.mills@uc.edu

Sourushe Zandvakili

Professor of Economics, University of Cincinnati

Phone: 513-254-3660 • Email: sourushe.zandvakili@uc.edu